

USB-C DAUGHTERBOARD / ENGINEERING DRAFT

Upstream: USB-C, USB 3.2 Gen 2 data UFP, USB-PD power source, 20 V / 5 A maximum.
Downstream: 2 Type-C + 3 Type-A SuperSpeed, 1 Type-A USB 2.0.
Use two GPU power inputs OR one EPS input. Not for simultaneous independent supplies.

3.3 V / 4 A – TPS56A37

File: Power_3V3.kicad_sch

USB7206C – six downstream ports

File: Hub.kicad_sch

Downstream Type-C 2 – 5 V / 3 A

File: USB_C2.kicad_sch

Downstream Type-A 3 – SuperSpeed / 900 mA

File: USB_A3.kicad_sch

Upstream Type-C – USB data UFP / PD source

File: USB_Upstream.kicad_sch

5 V / 12 A – TP540305

File: Power_5V.kicad_sch

Hub decoupling and transmit coupling

File: Hub_Decoupling.kicad_sch

Downstream Type-A 1 – SuperSpeed / 900 mA

File: USB_A1.kicad_sch

Downstream Type-A 4 – USB 2.0 / 500 mA

File: USB_A4.kicad_sch

STM32G071 – USB-PD policy and power control

File: PD_MCU.kicad_sch

1.15 V / 3 A – USB7206C core

File: Power_Core.kicad_sch

Downstream Type-C 1 – 5 V / 3 A

File: USB_C1.kicad_sch

Downstream Type-A 2 – SuperSpeed / 900 mA

File: USB_A2.kicad_sch

LM51772 – programmable 5–20 V / 100 W

File: PD_BuckBoost.kicad_sch

12 V input – two GPU 8-pin OR one EPS 8-pin

File: Power_Input.kicad_sch

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /
File: USB_C_Daughterboard.kicad_sch

Title: **USB C Daughterboard**

Size: A2	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 1/16

3.3 V / 4 A – TPS56A37

WEBENCH Design104 values. EN intentionally floating per datasheet section 6.3.5.
AGND/PGND join at regulator. Verify capacitance after DC-bias derating and inductor thermal limits.

A

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A

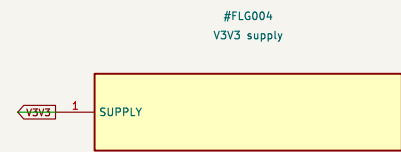
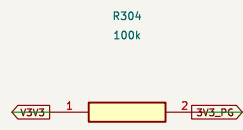
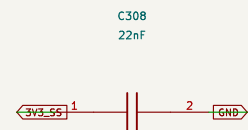
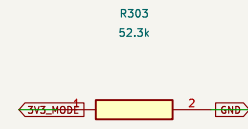
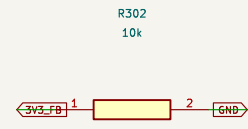
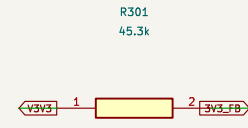
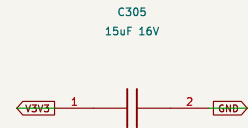
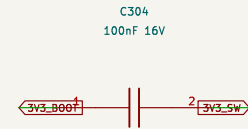
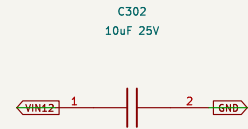
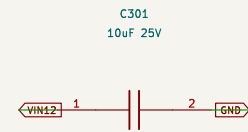
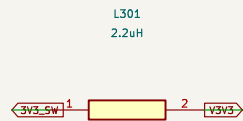
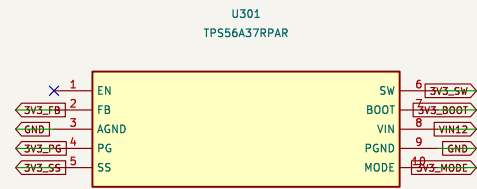
B

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F



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

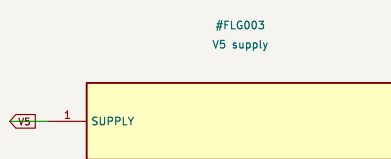
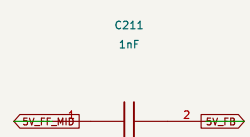
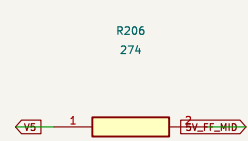
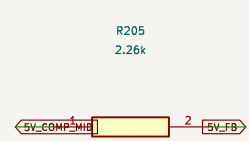
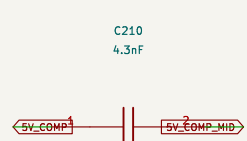
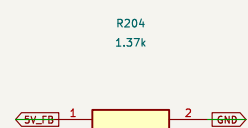
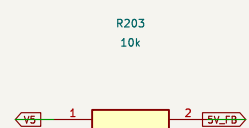
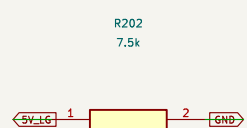
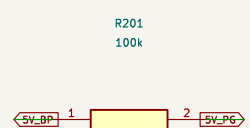
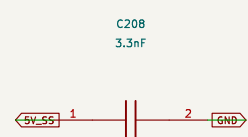
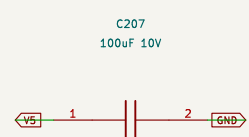
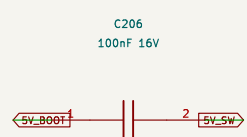
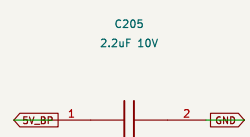
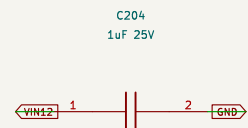
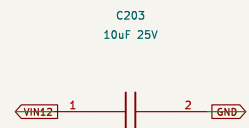
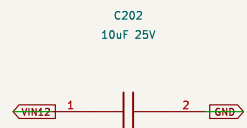
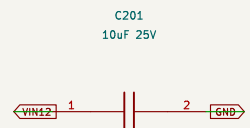
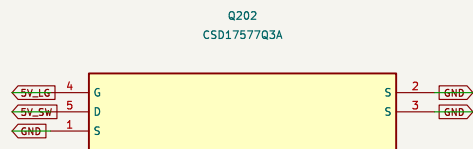
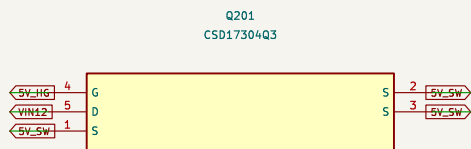
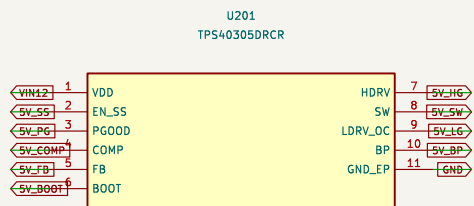
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Title: 3.3 V / 4 A – TPS56A37

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 2/16

5 V / 12 A – TPS40305

WEBENCH Design105 topology and values: 1.2 MHz. MOSFET drain pad numbering follows TI NexFET land pattern.
5 V allocation: two Type-C ports at 3 A, USB-A SDP loads, plus VCONN/mux overhead. BCL1.2 disabled.



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

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Size: A2	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 3/16

1.15 V / 3 A – USB7206C core

TPS62130, 12 V input. 0.8 V reference x (1 + 43.7k/100k) = 1.1496 V.
Hub VCore allowed range 1.09–1.21 V. Hold HUB_RESET_N low until both hub rails are stable.

A

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A

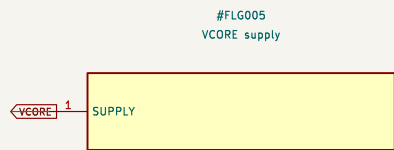
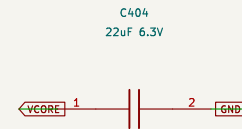
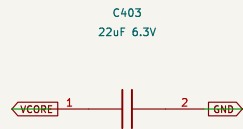
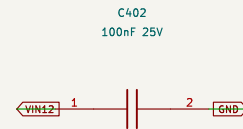
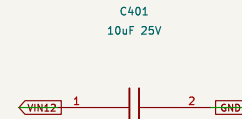
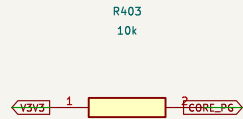
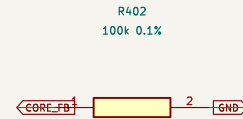
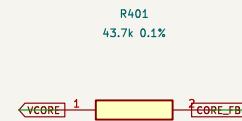
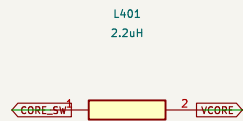
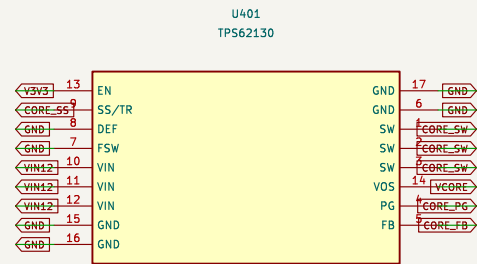
B

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F



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

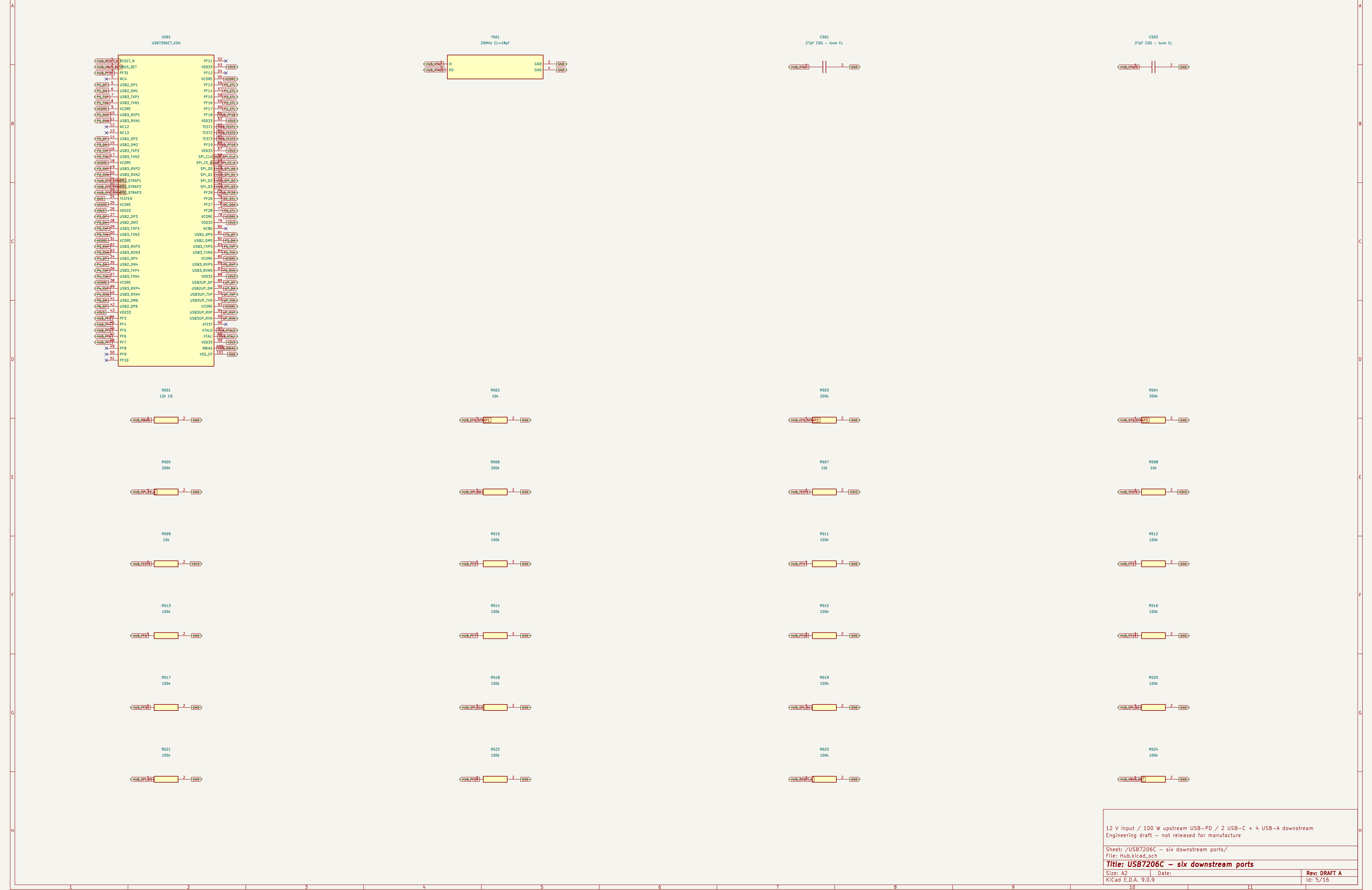
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Title: 1.15 V / 3 A – USB7206C core

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 4/16

USB7206C – six downstream ports

Port 1/2: Type-C. Ports 3/4/5: Type-A SuperSpeed. Port 6: Type-A USB 2.0.
Configuration 3 per D500003850F Tables 3–4/3–5. RESET and VBUS_DET driven by PD MCU, not high-voltage VBUS.



[illegible]

	1	2	3	4	5	6	7	8	9	10	11
A											
B											
C											
D											
E											
F											
G											
H											

Hub decoupling and transmit coupling

One 100 nF capacitor per VDD33/VCORE pin, placed at the relevant pin. Bulk capacitors near hub.
220 nF series capacitors on each hub SuperSpeed TX lane only. Route 90 ohm differential pairs without stubs.

C503 100nF	C504 100nF	C505 100nF	C506 100nF
C507 100nF	C508 100nF	C509 100nF	C510 100nF
C511 100nF	C512 100nF	C513 100nF	C514 100nF
C515 100nF	C516 100nF	C517 100nF	C518 100nF
C519 100nF	C520 100nF	C521 10uF 6.3V	C522 10uF 6.3V
C523 220nF 10V	C524 220nF 10V	C525 220nF 10V	C526 220nF 10V
C527 220nF 10V	C528 220nF 10V	C529 220nF 10V	C530 220nF 10V
C531 220nF 10V	C532 220nF 10V	C533 220nF 10V	C534 220nF 10V

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

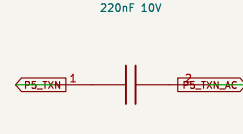
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Title: **Hub decoupling and transmit coupling**

Size: A2Date:

KiCad E.D.A. 9.0.9

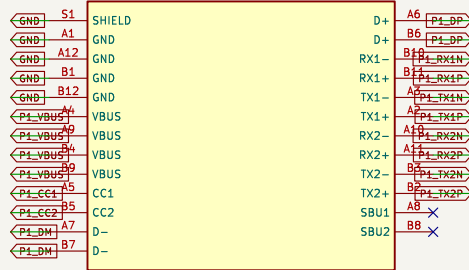
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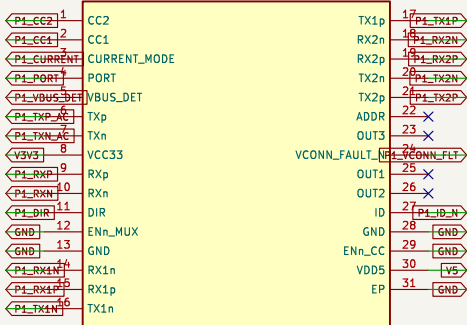
Downstream Type-C 1 – 5 V / 3 A

HD3SS3220 GPIO mode: ADDR open, PORT high, CURRENT_MODE 10k to 5 V.
VBUS enable = hub PRT_CTL AND attached, DIR has required 200k pull-up, eFuse nominal 3.27 A, latch-off.

U600
USB-C 24-pin 5A / 10Gbps



U600
HD3SS3220IRNHR



U602
74LVC1604



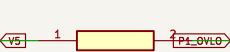
U603
74LVC1608



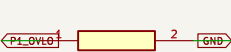
U601
TPS259470LRPW



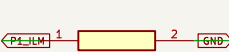
R601
36.5k



R602
10k



R603
1.02k 1X



C601
2.2nF



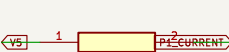
C602
1uF 10V



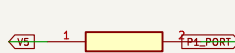
C603
10uF 10V



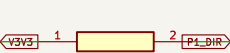
R604
10k



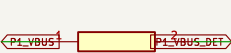
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10k



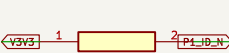
R606
200k



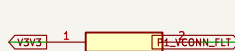
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900k



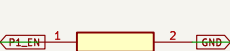
R608
10k



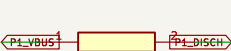
R609
10k



R610
100k



R611
330 0.25W



Q600
2N7002



C604
100nF



C605
100nF



C606
100nF



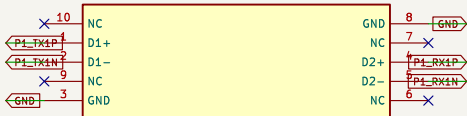
C607
100nF



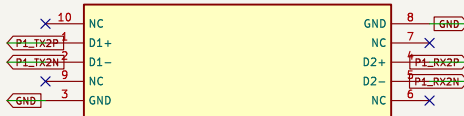
C608
4.7uF 10V



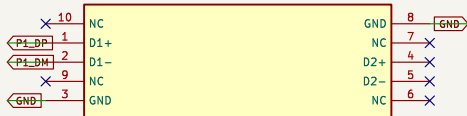
D600
TPD4E05U06DQA



D601
TPD4E05U06DQA



D602
TPD4E05U06DQA



D609
SMBJ5.0CA



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

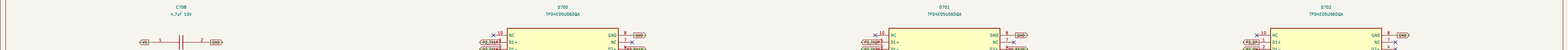
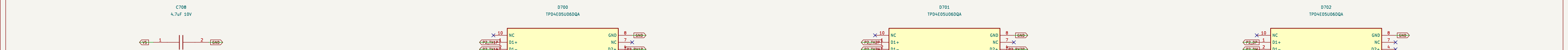
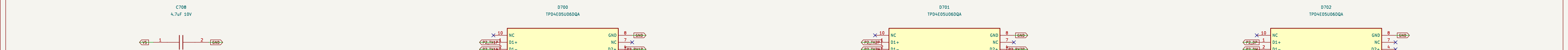
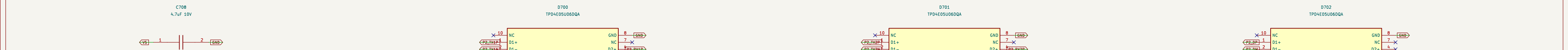
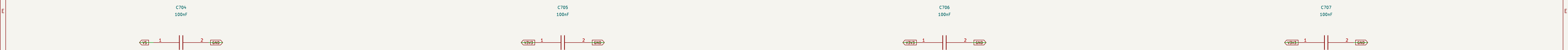
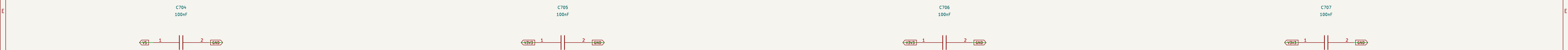
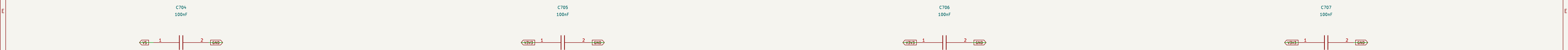
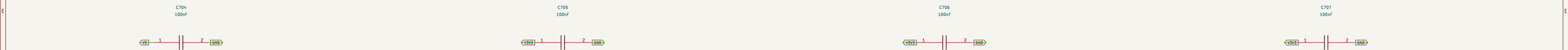
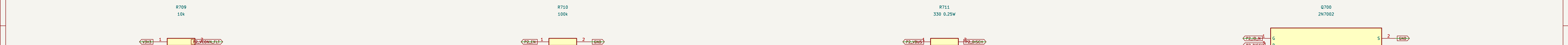
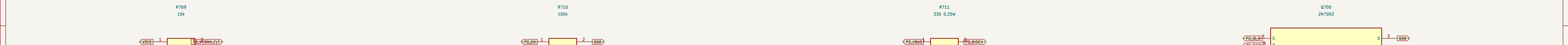
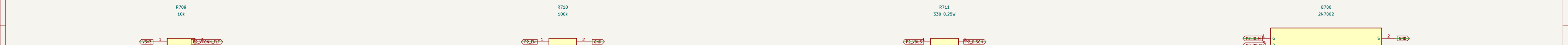
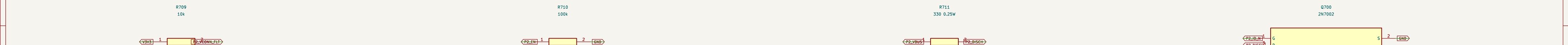
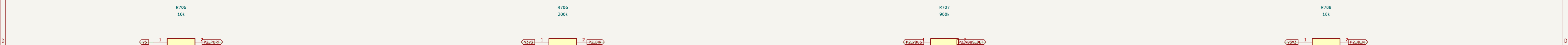
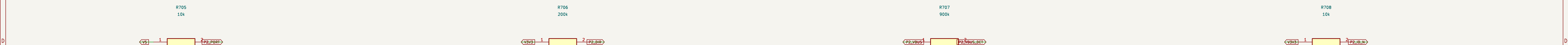
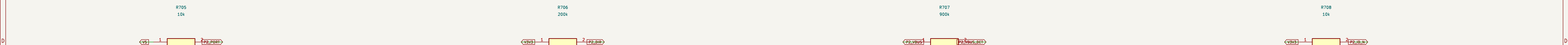
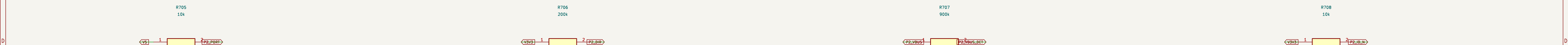
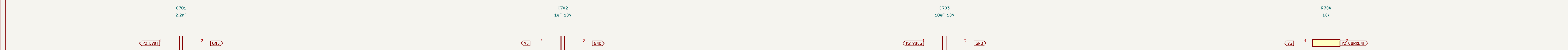
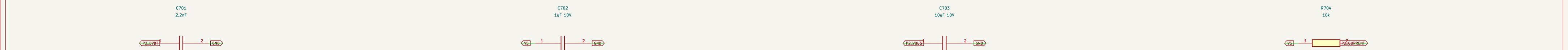
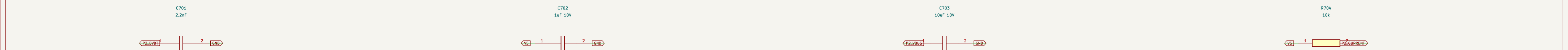
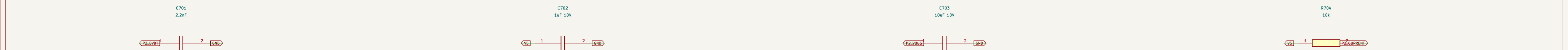
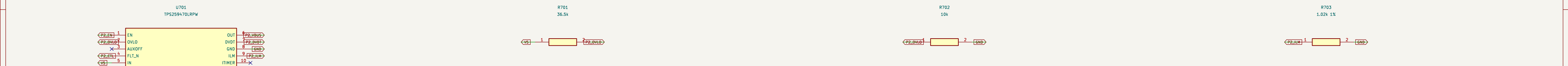
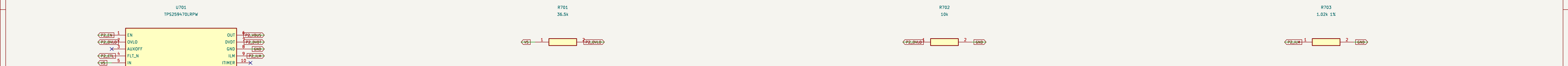
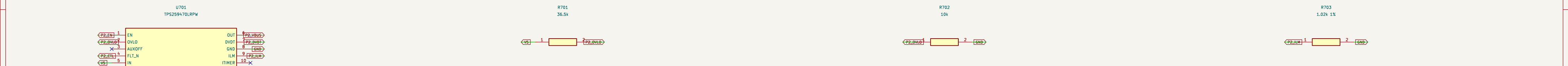
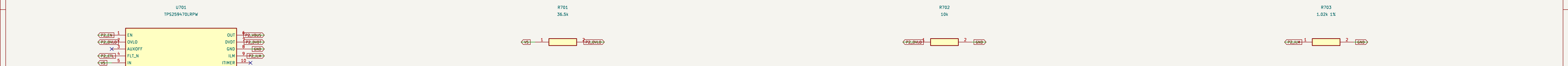
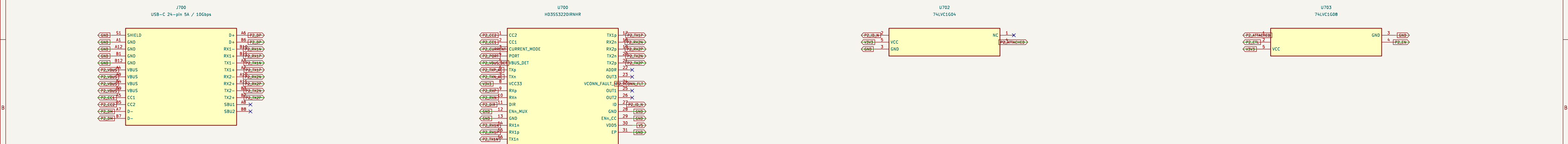
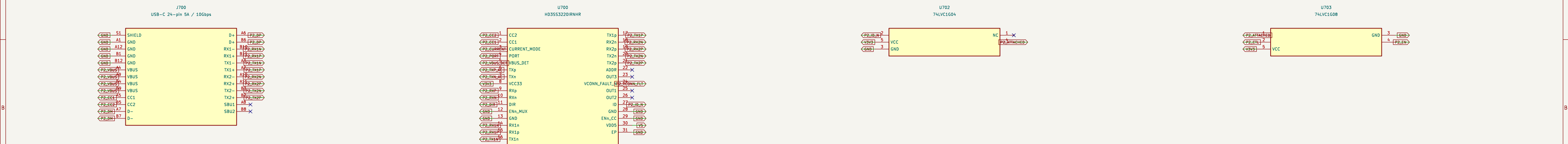
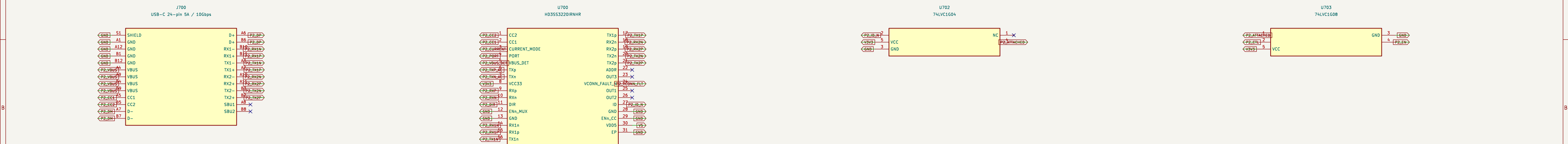
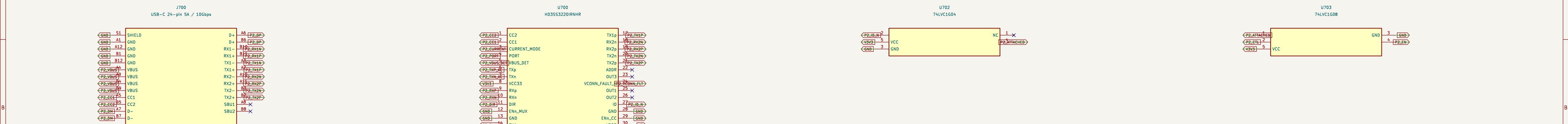
Sheet: /Downstream Type-C 1 – 5 V / 3 A/
File: USB_C1.kicad_sch

Title: Downstream Type-C 1 – 5 V / 3 A

Size: A2 Date: Rev: DRAFT A

KiCad E.D.A. 9.0.9 Id: 7/16

HD3SS3220 GPIO mode: ADDR open, PORT high, CURRENT_MODE 10k to 5 V.
VBUS enable = hub PRT_CTL AND attached. DIR has required 200k pull-up. eFuse nominal 3.27 A, latch-off.



Downstream Type-A 1 – SuperSpeed / 900 mA

TPS259470L provides per-port overcurrent protection and reverse-current blocking.
FLT_N returns to the hub combined PRT_CTL/overcurrent pin. BC1.2 disabled. Connector selection/footprint pending.

A

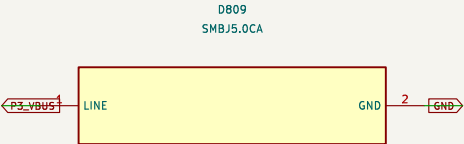
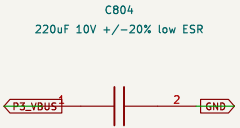
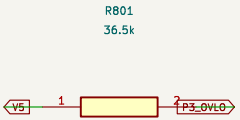
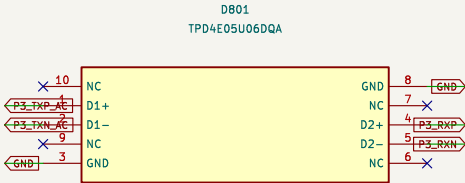
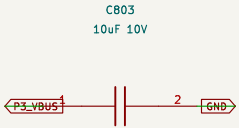
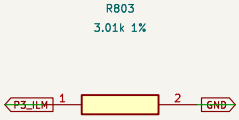
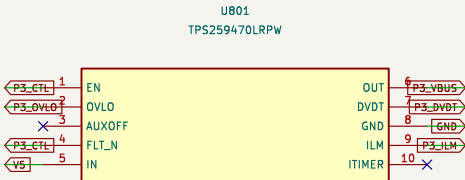
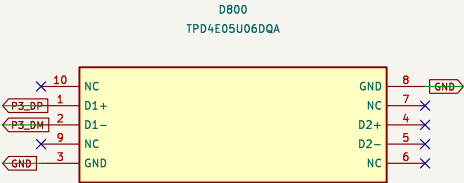
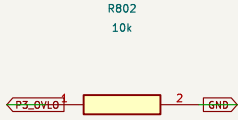
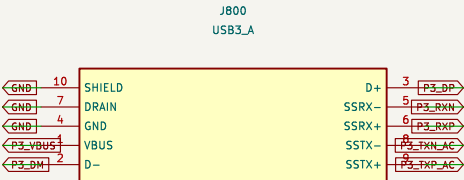
B

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12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /Downstream Type-A 1 – SuperSpeed / 900 mA/
File: USB_A1.kicad_sch

Title: Downstream Type-A 1 – SuperSpeed / 900 mA

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 9/16

Downstream Type-A 2 – SuperSpeed / 900 mA

TPS259470L provides per-port overcurrent protection and reverse-current blocking.
FLT_N returns to the hub combined PRT_CTL/overcurrent pin. BC1.2 disabled. Connector selection/footprint pending.

A

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A

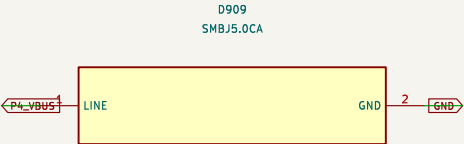
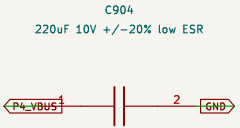
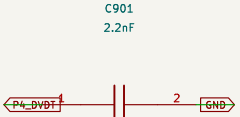
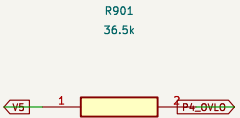
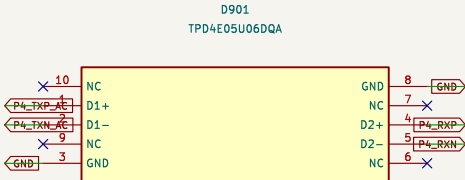
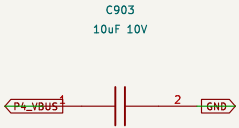
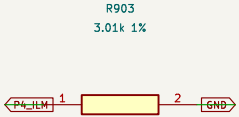
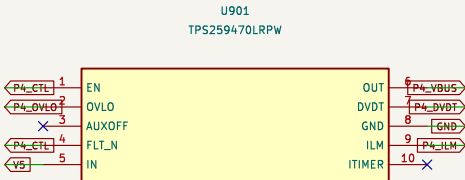
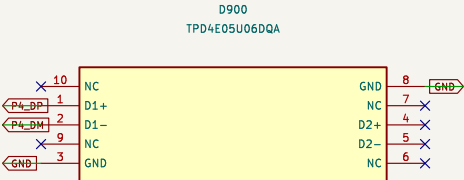
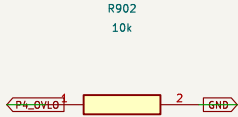
B

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F



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

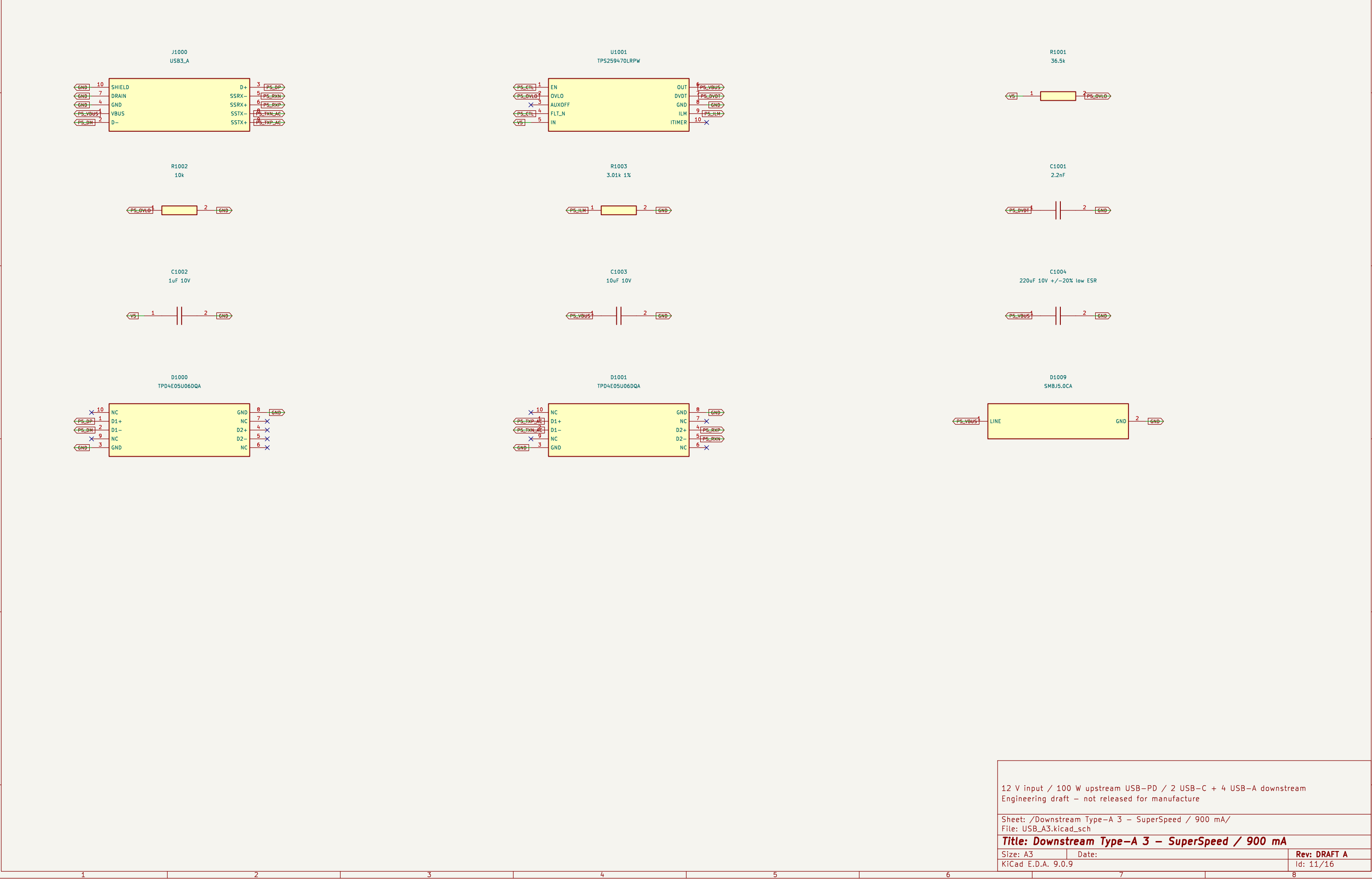
Sheet: /Downstream Type-A 2 – SuperSpeed / 900 mA/
File: USB_A2.kicad_sch

Title: Downstream Type-A 2 – SuperSpeed / 900 mA

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 10/16

Downstream Type-A 3 – SuperSpeed / 900 mA

TPS259470L provides per-port overcurrent protection and reverse-current blocking.
FLT_N returns to the hub combined PRT_CTL/overcurrent pin. BC1.2 disabled. Connector selection/footprint pending.



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /Downstream Type-A 3 – SuperSpeed / 900 mA/
File: USB_A3.kicad_sch

Title: Downstream Type-A 3 – SuperSpeed / 900 mA

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 11/16

Downstream Type-A 4 – USB 2.0 / 500 mA

TPS259470L provides per-port overcurrent protection and reverse-current blocking.
FLT_N returns to the hub combined PRT_CTL/overcurrent pin. BC1.2 disabled. Connector selection/footprint pending.

A

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A

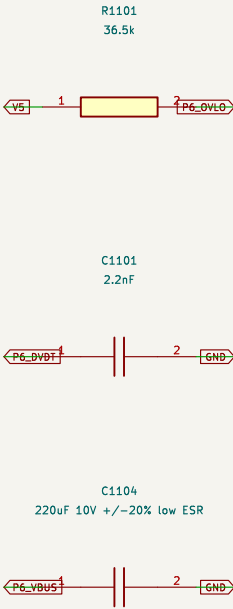
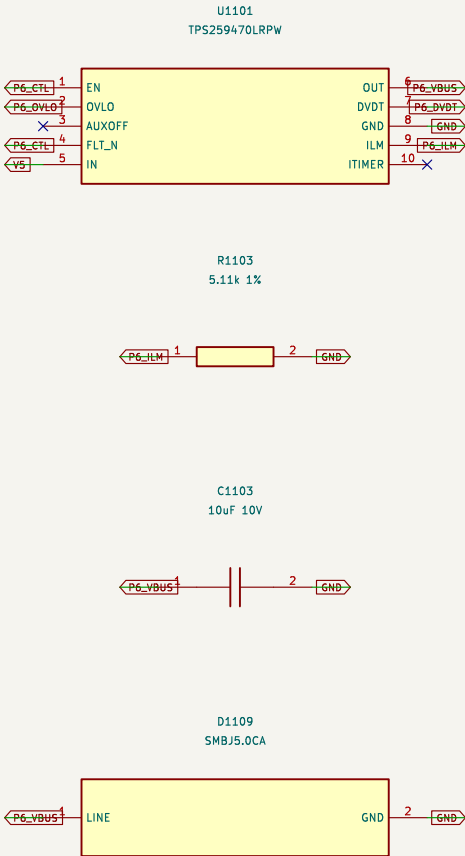
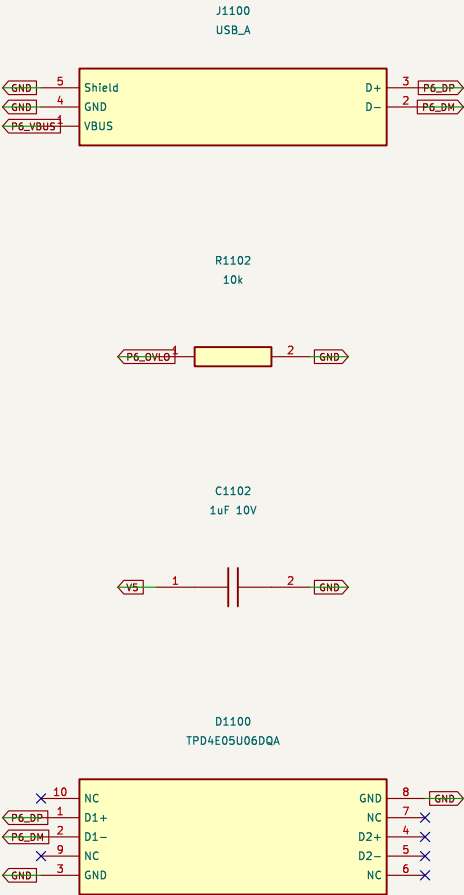
B

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12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

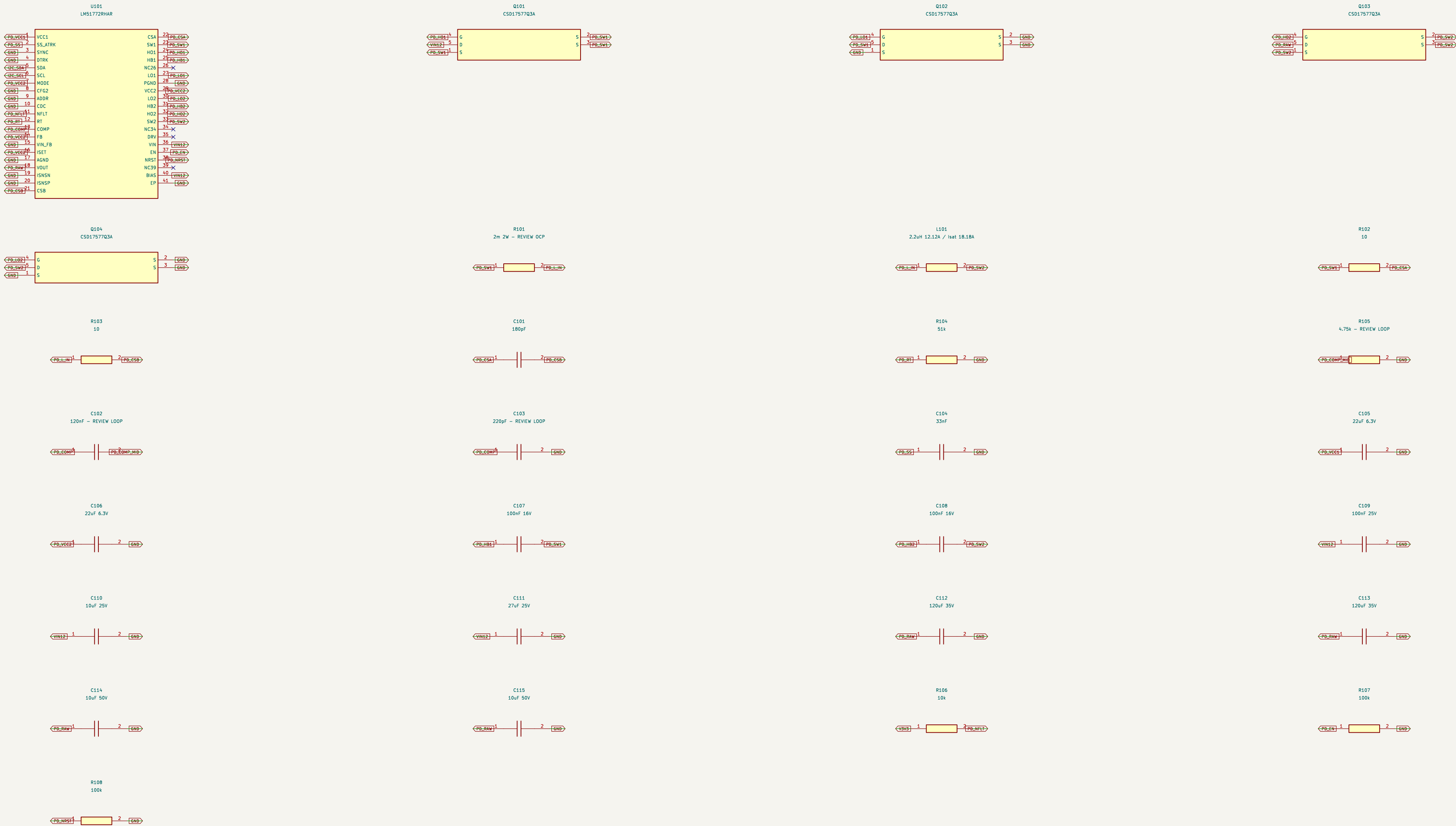
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File: USB_A4.kicad_sch

Title: Downstream Type-A 4 – USB 2.0 / 500 mA

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 12/16

LM51772 – programmable 5–20 V / 100 W

WEBENCH Design103 adapted for I2C (ADDR=GND, address 0x6A). Internal feedback and MCU enable/reset.
AP51Q4M2R2A replaces 1.8 uH. Compensation and current limit require recalculation/testing before power-up.
All bridge FETs standardized to CSD17577Q3A (30 V). Not an EPR design. Maximum contract 20 V / 5 A.



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /LM51772 – programmable 5–20 V / 100 W/
File: PD_BuckBoost.kicad_sch

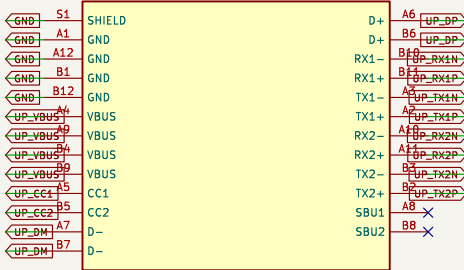
Title: LM51772 – programmable 5–20 V / 100 W

Size: A2	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 13/16

Upstream Type-C – USB data UFP / PD source

STM32 UCPD negotiates PD and manages data-role swap to UFP while keeping power-source role.
ICPP03 source path common-source back-to-back FETs. Sink path not filled. 7 mOhm gives nominal 6 A hardware OCP.
100 W requires identified 5 A e-marked cable; otherwise cap contract to 3 A. Hardware OVP celling -22 V.

U1200
USB-C 24-pin 5A / 100Gps



R1200
7m 1W 1%



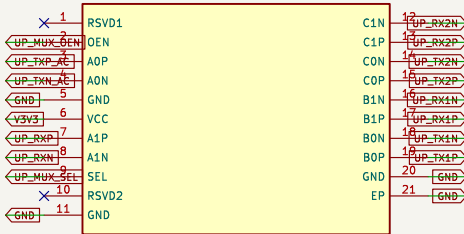
R1204
100k



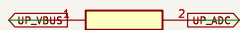
C1202
330pF 50V



U1201
HD3SS3212IRKSR



R1208
100k 0.1%



D1201
TPD4E05U06DQA



U1200
ICPP03-M20



R1201
10k 1%



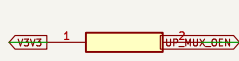
R1205
100k



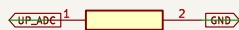
C1203
2.2uF 50V



R1206
100k



R1209
10k 0.1%



D1202
TPD4E05U06DQA



Q1200
CSD17577Q3A



R1202
560 1%



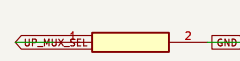
C1200
100nF 50V



C1204
100nF 10V



R1207
100k



C1207
1nF



D1203
5MBJ22CA



Q1201
CSD17577Q3A



R1203
10k



C1201
330pF 50V



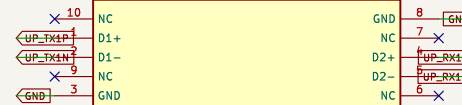
C1205
1uF 10V



C1206
100nF



D1200
TPD4E05U06DQA



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /Upstream Type-C – USB data UFP / PD source/
File: USB_Upstream.kicad_sch

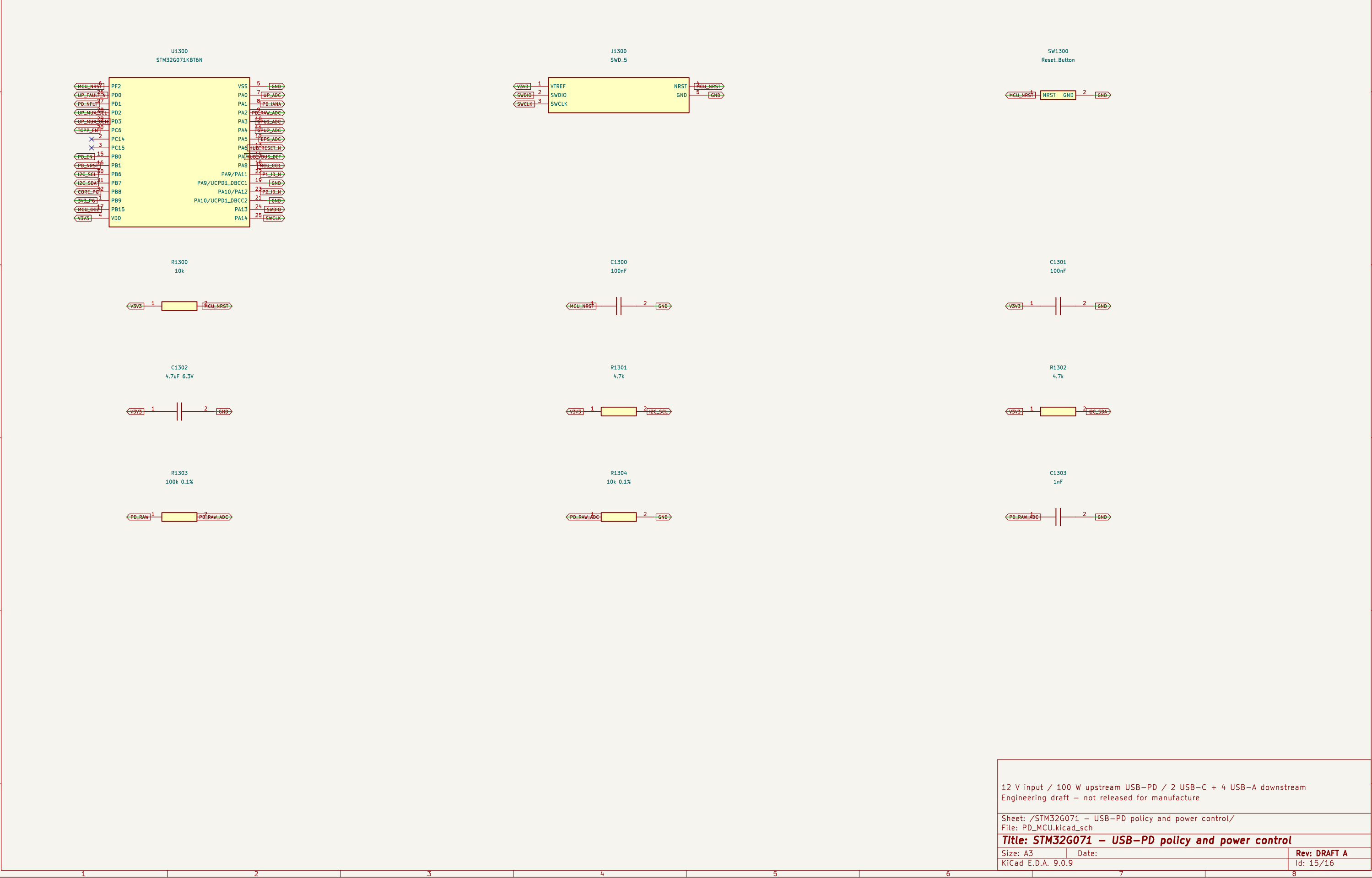
Title: Upstream Type-C – USB data UFP / PD source

Size: A2 Date: Rev: DRAFT A

KiCad E.D.A. 9.0.9 Id: 14/16

STM32G071 – USB-PD policy and power control

STM32G071KBT6N LQFP32: PA8=UCPD1_CC1, PB15=UCPD1_CC2, PB6/PB7=I2C1, PA13/PA14=SWD.
Source-only power path. UCPD dead-battery pins grounded; disable dead-battery function in firmware.
Boot from flash using option bytes; preserve NRST function. Firmware required; no PD stack is generated here.



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

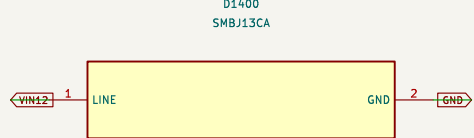
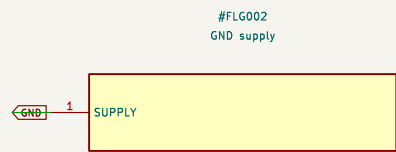
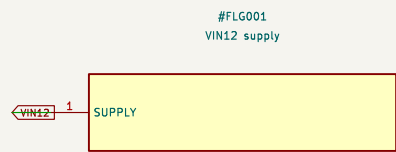
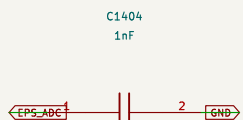
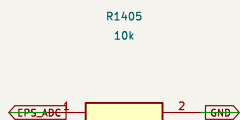
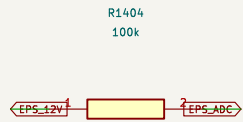
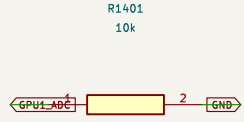
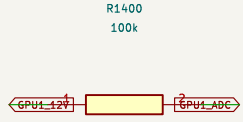
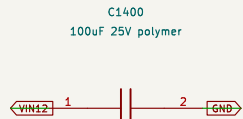
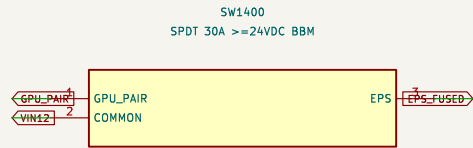
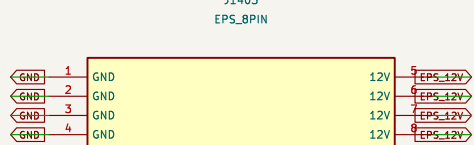
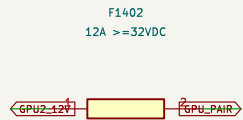
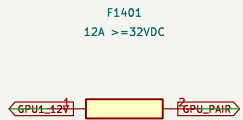
Sheet: /STM32G071 – USB-PD policy and power control/
File: PD_MCU.kicad_sch

Title: STM32G071 – USB-PD policy and power control

Size: A3	Date:	Rev: DRAFT A
KiCad E.D.A. 9.0.9		Id: 15/16

12 V input – two GPU 8–pin OR one EPS 8–pin

Choose GPU pair OR EPS with a break-before-make selector rated >=30 A, >=24 VDC. Do not switch under load.
GPU pair MUST come from the same PSU. Both GPU connectors required for full output. Do not combine separate PSUs.
Connector pin numbering is the electrical interface: verify keying, mating face and selected manufacturer drawing before layout.



12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /12 V input – two GPU 8–pin OR one EPS 8–pin/
File: Power_Input.kicad_sch

Title: 12 V input – two GPU 8–pin OR one EPS 8–pin

Size: A2 Date: Rev: DRAFT A
KiCad E.D.A. 9.0.9 Id: 16/16